HEALTHBRIDGE OVERVIEW REPORT

COMMUNITY HEALTH RESOURCE LOCATOR

Prepared by: Group 2

Institution: Mbarara University of Science and Technology

Course: Software Engineering

<u>PROJECT MEMBERS</u>

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EXECUTIVE SUMMARY

HealthBridge is a web-based platform that connects community members in underserved areas of Uganda to accurate, real-time information about nearby health clinics, medicine availability, and health education resources. The system replaces slow, error-prone manual processes (paper lists, Excel spreadsheets) with an interactive digital solution.

It supports three user roles Community Member, Clinic Administrator, and HealthBridge Administrator each with tailored dashboards.

Privacy for sensitive health searches (HIV, mental health, STDs) is a core design principle. The system includes an interactive map (Leaflet.js), role-based access control, clinic management, and administrative reporting.

SECTION 1: INTRODUCTION

1. **Problem Statement**

   In many underserved Ugandan communities, patients and healthcare workers rely on outdated, unreliable methods to locate health services. Clinic lists are often kept on paper notice boards or static Excel files. When a patient needs a specific service (e.g., HIV clinic, maternity care, or a particular medicine), a social worker must manually search through these records, call clinics to verify availability, and then call back the patient – a process that can take hours or days. This leads to:

   - Delayed access to care
   - Outdated or incorrect information
   - Wasted staff time
   - Privacy risks for sensitive health queries

2. **Proposed Solution**

HealthBridge provides a centralised, web based platform where;

a) Community members can search for clinics by location, service type, or medicine availability, view results on an interactive map, and read health articles.
b) Clinic administrators can update their clinic's hours, contact details, and medicine stock.
c) HealthBridge administrators can add new clinics, verify data, generate reports, and manage health articles.
d) All sensitive searches (HIV, mental health, STDs) are protected by explicit user consent and are never logged, preserving patient privacy.

3. **Objectives**

- ✓ Reduce the time to find a health resource from hours to under 2 minutes.
- ✓ Provide 100% verified clinic data (verified at least every 90 days).
- ✓ Ensure that sensitive health searches are never stored or logged.
- ✓ Enable non-technical clinic staff to update their own information without developer help.
- ✓ Deliver a responsive, map-based interface accessible on low-end smartphones.

SECTION 2: SYSTEM OVERVIEW

1. **Target Users**

Role	Description	Key Tasks
Community Member	Patient or individual seeking health services	Search clinics, filter by service/medicine, view map, read articles
Clinic Administrator	Staff at a specific clinic	Update clinic hours, contact info, medicine stock
HealthBridge Administrator	HealthBridge coordinator	Verify clinics, generate reports,

		manage articles, oversee system

2. **Core Features**
   - Role-based authentication (login/registration).
   - Clinic search by district, service type, or medicine name. Interactive map (Leaflet.js / Open Street Map) showing clinic locations.
   - Privacy protection for sensitive searches (consent pop-up, no logging).
   - Clinic admin panel, edit own clinic, add services, update medicine stock.
   - Super admin panel, verify clinics, generate usage reports, manage health articles.
   - Health articles, educational content with category filtering.

3. **Technology Stack (as per Architectural Design)**

Layer	Technology	Purpose
Frontend	React.js, Leaflet.js	User interface, maps, responsive styling
Backend	Node.js, Express.js	REST API, business logic, authentication
Database	PostgreSQL	Relational data (users, clinics, medicines)
Caching	Redis	Speed up frequent searches
Deployment	Docker	Scalable hosting
Version Control	GitHub	Scalable hosting

Note: The prototype uses a simplified stack (HTML/CSS/JS + localStorage) to demonstrate functionality without installation. The production system will use the full stack above

SECTION 3: SYSTEM ARCHITECTURE

1. **High-Level Architecture (MVC + Layered)**
2. **Key Components**

Component	Responsibility
Search Engine	Processes location/service/medicine queries, returns ranked clinics
Map Module	Displays clinic markers using Leaflet.js and Open Street Map
Auth Module	Manages JWT tokens, role-based access control (RBAC)
Admin Panel	CRUD operations for clinics, verification, reports
Medicine Inventory	Tracks stock status (In Stock / Low Stock / Out of Stock) per clinic
Privacy Manager	Detects sensitive search terms, shows consent pop-up, prevents logging.

SECTION 4: IMPLEMENTATION (PROTOTYPE)

1. **Prototype Features**

The working prototype (HTML/CSS/JS) implements:

1) Login / Registration – users stored in browser local Storage
2) Three role dashboards;
    - Community: search, map, health articles
    - Clinic Admin: edit own clinic, update stock, add services
    - HealthBridge Admin: verify clinics, generate reports, manage articles
3) Interactive map (Leaflet) with clinic markers
4) Privacy protection – pop-up consent before sensitive searches
5) Location-based search – “Use my location” button

2. **Known Limitations (Prototype)**

- Data is stored in localStorage (not a real database)
- No backend API (client-side only)
- Passwords are plain text (no hashing)
- No real-time updates across users
- Map uses OpenStreetMap (no Google Maps API key required)

3. **<u>Testing Feedback (from Counterpart Group)</u>**

   The following comments were collected during peer testing;

   - Slow response under heavy load, The system showed delays when multiple users accessed it simultaneously.
   - SQL injection vulnerability, Input fields were not fully protected against malicious SQL attacks.
   - Outdated clinic data displayed, Some clinic information shown to users was not current.
   - Map fails on low internet, The interactive map did not load properly on weak connections.
   - Broken links not handled, Invalid links caused error pages instead of user-friendly messages.
   - Pharmacy search returned wrong results, Searching for pharmacies showed other facility types.
   - Empty search not properly validated, Blank searches did not always prompt the user to enter data.
   - Invalid credentials sometimes granted access, Login with wrong passwords occasionally worked.
   - Fast multiple searches caused delays, Rapid consecutive searches slowed down the system.

SECTION 5: DEPLOYMENT PLAN (Production)

1. **<u>Environment</u>**

Component	Technology	Hosting
Frontend	React.js (build)	Firebase Hosting / Vercel
Backend	Node.js + Express	Google Cloud Run (container)
Database	PostgreSQL	Google Cloud SQL
Cache	Redis	Google Cloud Memorystore
Static assets	Images, CSS	Cloud CDN
Domain	SSL (HTTPS)	Custom domain (for example healthbridge.ug)

2. **Deployment Steps (Future Work)**
    a. Set up PostgreSQL schema (from schema.sql)
    b. Deploy backend to Cloud Run with environment variables
    c. Deploy frontend build to Firebase Hosting
    d. Configure custom domain and SSL certificate
    e. Set up automated backups (daily)
    f. Implement monitoring (uptime, error alerts)

SECTION 6: FUTURE ENHANCEMENTS (Based on Feedback)

I. Implement input validation and encryption, Protect all user inputs against SQL injection and XSS attacks.
II. Optimise database queries and add caching, Use Redis to cache frequent searches and improve response time under load.
III. Regular data update mechanism, Automated weekly verification reminders for clinic admins to keep data current.

IV. Offline map fallback, Display a static map or text-based clinic list when internet connection is poor.

V. Improve error handling, Show user-friendly messages for broken links, empty searches, and failed operations.

VI. Fix search filters, Ensure pharmacy, maternity, and other service filters return correct clinic types.

SAMPLE CLINICS INCLUDED

Clinic Name	District	Services
Ruharo Mission Hospital	Mbarara	Maternity, General Consultation, Pharmacy, HIV/AIDS
TASO Mbarara	Mbarara	HIV/AIDS, Pharmacy, Counselling
Kampala General Hospital	Kampala	Maternity, Dental, Emergency
Nsambya Hospital	Kampala	Maternity, Paediatrics
Gulu Regional Hospital	Gulu	General Consultation, Surgery, HIV/AIDS
Jinja Regional Hospital	Jinja	Maternity, General Consultation
Mental Health Center – Butabika	Kampala	Mental Health, Counselling

CONCLUSION

HealthBridge successfully demonstrates how a well-designed, privacy-aware web application can solve the real-world problem of accessing reliable health resource information in underserved Ugandan communities. The prototype validates all core user journeys, from community member search to clinic admin updates to super admin verification while upholding strong privacy principles for sensitive health queries.

The architectural design (MVC + layered, 4+1 views, client-server) is sound and ready for implementation using the documented technology stack (React, Node.js, PostgreSQL, Redis). Feedback from peer testing has been incorporated into the future enhancements list, ensuring that the next iteration will be even more robust, accessible, and scalable.

HealthBridge is not just a student project it but rather has the potential to be deployed in real NGOs and health centres, improving health outcomes for thousands of Ugandans.